JOINT ANGLE MEASUREMENT USING MARKER-BASED OPTICAL MOTION CAPTURE DURING SUBMAXIMAL AND MAXIMAL BROAD JUMPING

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INTRODUCTION

Lower limb injuries such as anterior cruciate ligament (ACL) tears are one of the most common and expensive sports related injuries. This mainly happens in sports where jumping and landing is most common such as soccer, basketball, volleyball, or American football. Athletes mainly in high-risk sports such as these experience these ACL injuries due to poor neuromuscular control and increased loading during landing [1]. Dynamic joint stabilization depends on the balance between a muscle force and passive ligament restraints. When this control fails, there's a lot of 3D forces transferred to the knee which increases the risk of ligament damage. This would show why studying joint kinematics during jump movements is important.

The objective of this lab was to figure out whether the hip, knee, or ankle exhibits a bigger range of motion (ROM) in the sagittal plane and faster angular velocity (AV) using both single (SL) and double leg (DL) broad jumps at different effort levels. It was hypothesized that the hip joint would show the biggest range of motion and angular velocity at 100% effort due to its proximal position and role in initiating the extension of the knee and ankle to generate maximal power.

METHODS

One participant (male, age 20, 66kg, and 1.795m in height) provided written consent before conducting any experimental testing approved by the University of Guelph Research Ethics Board. This participant wore T-shirt, sweatpants, and running shoes. Participant completed 1-2 trials to understand experimental procedure, keep consistency, and reduce distortion in the data from the soft tissue artifacts. Three-dimensional (3D) data was collected using the Vicon Motion Capture System (v2.16 Vicon Motion Systems Ltd., Oxford, UK) with a sampling rate of 200Hz throughout the experiment. Ensured Kistler Piezoelectric Force Plate (9281EA, Kistler AG, Winterthur, Switzerland) was operational and sampling rate is 1000 Hz or greater. Mocap reflective markers were attached to the participant with double sided tape keeping the bottom side of the marker on double sided tape to prevent motion artifacts. Markers were placed on participant following the HASMotion "Coda Pelvis", "Thigh Model 1", "Shank Model 1" (without RTAM), and "Multi Segment Foot" placements (HASMotion Inc., Kingston, ON, Canada). First trial consisted of a static calibration trial recorded in Vicon Nexus (v2.16) to define the anatomical axes for each body segment based on the 3D positions of the markers. This trial was also done to confirm that the markers are all fully visible before the dynamic trials were done.

Participant performed two broad jumps with one being with submaximal effort and the other maximal effort. Jumps were done from a standing start with hands on hips and a 30 second rest to prevent fatigue. Submaximal trial was used to confirm the tracking accuracy, and the maximal trial was considered the primary condition. Both landing and take-off had to occur within the data collection. Measured jump distances included 2.41m for the DL jump and 1.76 m for the SL jump. Jump efforts for both DL and SL were analyzed (100%, 75%, 50%, 25%). Mocap marker trajectories were reconstructed, had gaps filled using pattern fill, and labelled each marker according to HASMotion placements (HASMotion Inc., Kingston, ON, Canada) on Vicon Nexus. Processed values were imported to Visual 3D (v2023, C-Motion Inc., MD, USA) for modelling. Linked segment models were made according to ISB standards using a Caudan X-Y-Z rotation sequence. The marker trajectories were filtered at 6 Hz and hip, knee, and ankle joint angles were calculated based off their proximal segments. Lastly, the final data was plotted using MATLAB (R2025a, MathWorks Inc, Natick, United States) to create Fig.1 and Table 1.

RESULTS

Both peak angular velocity (AV) and joint range of motion (ROM) increased with higher efforts for all joints. Knees showed the biggest sagittal ROM and AV at high effort levels from DL75-100 (140.22° & 1702.5°/s) (Table 1). Same goes for the hip (79.094° & 523.97°/s) and ankle (61.133° & 1631°/s) (Table 1). The DL jumps generally produced bigger values compared to the SL jumps.

All adduction angles increased according to the effort levels but remained below 30° (Fig. 1). Largest angles were observed for the 100% effort DL jump for the knee proving the ROM and AV increased with the effort levels. Lastly, the hip showed the most consistent increase across effort levels.

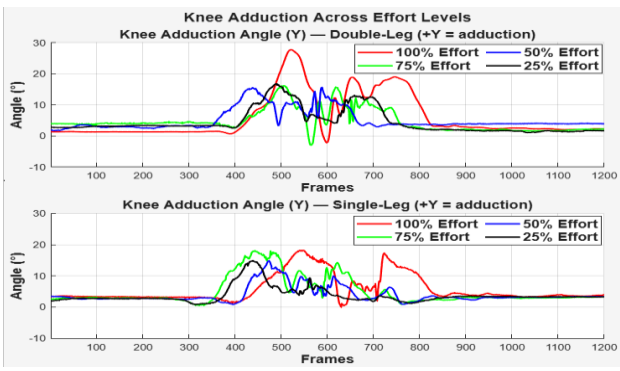

Figure 1: Knee adduction angles (°) across effort levels (100%, 75%, 50%, 25%) for DL (top) and SL (bottom) jumps.

DISCUSSION

The hip, knee, and ankle all show increased values with higher effort levels with the knee having the largest sagittal ROM and AV (Table 1). Since the hip did not have the highest values at 100% effort level, the hypothesis was not met.

The knee contributed the most to the broad jump distance since it has a higher ROM and AV, and the hip acted as a power generator. According to Research Methods in Biomechanics, the lower limbs have individual muscles that produce forces that cause resultant joint movements [2]. This shows the jump distance is found by the combined joint movements from the hip and knee. This means the hip generates power for the knee extension which resulted in bigger ROM, AV, and jump distance.

Knee adduction angles increased with effort levels with the peak at the 100% DL jump but remained below 30° (Fig. 1). The broad jump screening test may be used to determine knee injuries by discovering the amount of knee adduction or asymmetry during landing. Although, there are better ways to test knee injuries such as an EMG analysis.

Key limitations from this study were the marker placement variability and soft tissue artifact which affected the calculated joint angles. More accurate methods of tracking can improve the accuracy of the results.

CONCLUSION

The results did not support the hypothesis, instead proved that while the knee exhibited the biggest ROM and AV for the DL and SL jumps. In the future to reduce minor errors in joint angle accuracy from marker placement variability and soft tissue artifact, experiment should use rigid marker clusters since they take the average for all markers on a segment making it more accurate.

REFERENCES

1. Hewett TE, Myer GD, Ford KR, Heidt RS, Colosimo AJ, McLean SG, Van Den Bogert AJ, Paterno MV, Succop P. Biomechanical Measures of Neuromuscular Control and Valgus Loading of the Knee Predict Anterior Cruciate Ligament Injury Risk in Female Athletes: A Prospective Study. Am J Sports Med. 2005 Apr;33(4):492–501.

2. Robertson DGE, Caldwell GE, Hamill J, Kamen G, Whittlesey SN. Research Methods in Biomechanics. 2nd ed. Human Kinetics; 2014.

Table 1: Mean sagittal plane ROM (°) and peak AV (°/s) for hip, knee, and ankle joints across effort levels (100%, 75%, 50%, and 25%) and jump type (SL /DL).

Joints	Range of Motion (°)								Peak Angular Velocity (°/s)							
	DL100	SL100	DL25	SL25	DL50	SL50	DL75	SL75	DL100	SL100	DL25	SL25	DL50	SL50	DL75	SL75
Hip	79.094	61.289	63.186	40.678	64.888	47.214	69.791	53.205	434.91	375.77	339.49	212.34	413.41	270.56	523.97	486.19
Knee	140.22	79.482	69.104	53.718	82.848	63.525	95.731	66.345	889.82	1136	654.07	492.31	1055.9	596.24	1702.5	862.26
Ankle	61.133	56.284	52.163	54.135	55.446	47.807	60.373	59.769	589.17	1400	1394.3	935.53	1343	1210.6	1631	1571.9